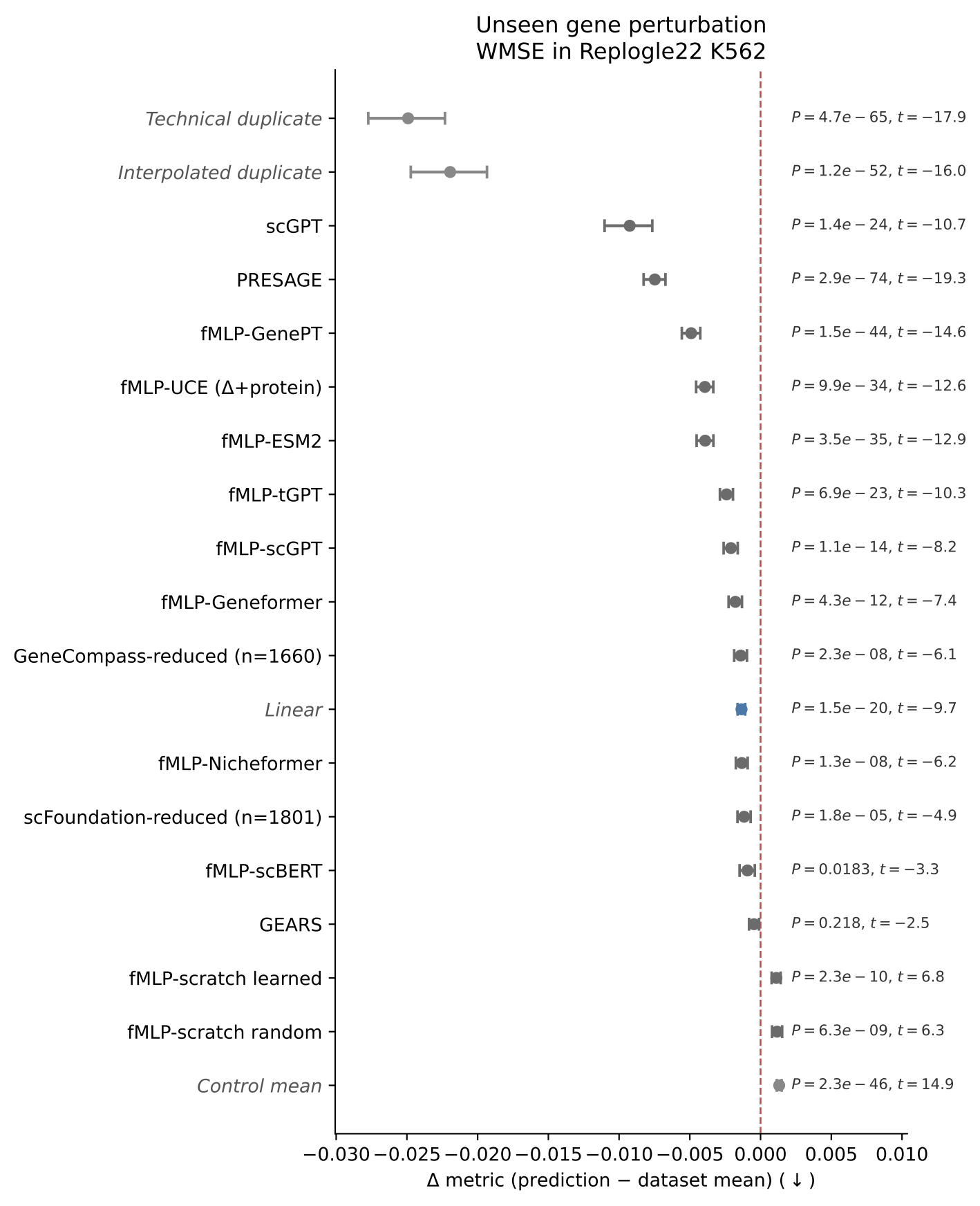
Unseen gene perturbation
WMSE in Replogle22 K562
Technical duplicate
Interpolated duplicate
scGPT
PRESAGE
fMLP-GenePT
fMLP-UCE (Δ+protein)
fMLP-ESM2
fMLP-tGPT
fMLP-scGPT
fMLP-Geneformer
GeneCompass-reduced (n=1660)
Linear
fMLP-Nicheformer
scFoundation-reduced (n=1801)
fMLP-scBERT
GEARS
fMLP-scratch learned
fMLP-scratch random
Control mean
P = 4.7e − 65, t = −17.9
P = 1.2e − 52, t = −16.0
P = 1.4e − 24, t = −10.7
P = 2.9e − 74, t = −19.3
P = 1.5e − 44, t = −14.6
P = 9.9e − 34, t = −12.6
P = 3.5e − 35, t = −12.9
P = 6.9e − 23, t = −10.3
P = 1.1e − 14, t = −8.2
P = 4.3e − 12, t = −7.4
P = 2.3e − 08, t = −6.1
P = 1.5e − 20, t = −9.7
P = 1.3e − 08, t = −6.2
P = 1.8e − 05, t = −4.9
P = 0.0183, t = −3.3
P = 0.218, t = −2.5
P = 2.3e − 10, t = 6.8
P = 6.3e − 09, t = 6.3
P = 2.3e − 46, t = 14.9
−0.030 −0.025 −0.020 −0.015 −0.010 −0.005 0.000 0.005 0.010
Δ metric (prediction − dataset mean) (↓)